

Research on the improvement of the marching cubes algorithm for three-dimensional reconstruction and visualization system of tumor target areas

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Abstract—Three-dimensional reconstruction technology can present the three-dimensional structure of tumor target areas in a stereoscopic manner, providing visual support for the formulation of radiotherapy plans. Due to its simplicity and efficiency, the Marching Cubes algorithm has become the mainstream isosurface extraction method in medical image three-dimensional reconstruction. To address issues such as hole artifacts and redundant computations in empty voxels associated with traditional algorithms, this study proposes an improved MC algorithm: it reduces empty voxel computations through surface connectivity theory and optimizes floating-point operations using the midpoint selection method. Experiments show that the improved algorithm reduces the number of triangular facets by more than 4% while improving processing efficiency by over 16%, ensuring reconstruction quality. This advancement provides a new solution for real-time three-dimensional visualization of tumor target areas.

Keywords—Marching Cubes, tumor target area, 3D reconstruction, visualization

I. INTRODUCTION

As a major challenge in the field of global public health, cancer has become one of the leading causes of death in most countries. According to statistics, the number of deaths caused by cancer accounts for nearly one-sixth of the total global deaths, and it accounts for as high as 22.8% of the deaths related to non-communicable diseases[1]. Radiotherapy, is a cancer treatment method that utilizes high-energy rays or particles to kill cancer cells or shrink tumors. It prevents cancer cells from dividing and growing by damaging their DNA, so as to achieve the goal of controlling or eliminating tumors. In radiation therapy, accurately defining the target area delineated within the irradiation scope is a crucial step for precise treatment. Doctors can intuitively observe the boundaries of the tumor and its specific location within the body by viewing the three-dimensional model. The 3D model can display the complex shapes and irregular edges of the tumor, helping doctors identify details that might be missed in two-dimensional images. This ensures the accuracy and completeness of target area delineation, laying a solid foundation for precise radiotherapy.

The MC (Marching Cubes) [2] algorithm, proposed by Lorensen and Cline in 1987, is a classic algorithm for isosurface extraction in three-dimensional data fields. This algorithm processes 3D data in the order of scan lines and uses linear interpolation to calculate the vertices of triangles. Due to its good three-dimensional reconstruction effect, simple

algorithm principle characteristics, it is especially suitable for medical image processing [3]. However, the traditional MC algorithm still has some drawbacks, such as its ambiguous topological structure and long computation time, etc [4]. In 2003, Lopes and Brodlie proposed a new improved algorithm [5]. By using linear interpolation to calculate the coordinates of intersection points and determining equivalent patterns, they simplified the algorithm, improved computational efficiency, and ensured the accuracy of surface representation. However, this algorithm does not take into account all cases involving two saddle points within a voxel. In 2011, Vignoles and Donias proposed the SMC (Simplified Marching Cubes) algorithm to address the issues of the non-smooth nature of the reconstructed models and the excessive number of polygons generated by the MC algorithm [6]. This algorithm uses the vertices of voxels instead of interpolating the intersection points between voxels. This method effectively avoids the ambiguity problem of the MC algorithm. At the same time, the number of triangular isosurfaces generated is significantly reduced, thus improving the reconstruction efficiency and simplifying the model. However, the accuracy of the SMC algorithm is not as good as that of the MC algorithm. In 2021, Xin WANG and others proposed an MC algorithm based on edge growth [7]. This algorithm can effectively solve the problem of holes caused by the ambiguity issue and can reconstruct the outer contour of the model with high quality. However, this algorithm has a slow calculation speed, requires more memory, and has relatively high requirements for the performance of computer hardware.

In summary, the traditional MC algorithm cannot solve the following problems simultaneously. Firstly, there is the issue of ambiguity. The isosurfaces between voxels have topological mismatches, resulting in a large number of holes in the reconstructed outer contour. Secondly, the reconstruction calculation time is long. The traditional algorithm needs to traverse all voxels. Since there are a large number of empty voxels in the scalar field data of the target area, that is, voxels that do not intersect with the isosurface, the algorithm runs slowly. Thirdly, when the isosurface of the MC algorithm is close to the nodes, it is prone to generate bad triangles, that is, two or more triangles share a short side, forming a large number of useless and elongated triangles. Fourthly, the reconstructed model is not smooth and there is a large number of polygons.

To address the aforementioned issues simultaneously, this paper proposes an improved MC algorithm based on the theory

of surface connectivity within volumetric data. By selecting a seed triangle as the starting point of the isosurface and progressively expanding this seed triangle through the theory of surface connectivity within volumetric data, the entire isosurface of the 3D model is generated. This method effectively resolves the issue of holes and circumvents unnecessary computations on empty voxels. Moreover, this paper adopts the midpoint selection technique in lieu of the traditional linear interpolation method, which not only significantly reduces computational load but also avoids the generation of minuscule triangles. As a result, it further accelerates the speed of three-dimensional reconstruction and renders the contours of the reconstructed models smoother.

The visualization system for tumor target areas is a pivotal technology in modern radiotherapy. It transforms medical imaging data into intuitive three-dimensional models, aiding physicians in more accurately delineating the boundaries of tumors and surrounding tissues, and in devising optimal radiotherapy plans.

II. IMPROVED MC ALGORITHM

The MC algorithm is a classical method for extracting isosurfaces from three-dimensional volumetric data. It partitions the volumetric data into cubic units, known as voxels. By comparing the values at the eight vertices of each voxel with a preset threshold, it determines whether the isosurface intersects the voxel and generates corresponding triangular facets, thereby achieving efficient reconstruction of three-dimensional surfaces. It achieves reconstruction by dividing the volumetric data into minute voxels and implementing isosurface extraction within each voxel. Specifically, the algorithm determines the intersection points of the isosurface with the voxel boundaries and employs linear interpolation techniques to ascertain the precise locations of these intersections. By comparing the values at each voxel vertex with a preset threshold, it deduces whether the isosurface resides within that voxel. Depending on the varying combinations of vertex values, a voxel can be decomposed into a maximum of five triangles, with the organization of these triangles contingent upon the intersection of the isosurface with the cubic grid. Since each vertex can exist in one of two states, a cubic grid comprising eight vertices has a total of $2^8=256$ possible state combinations. Through the analysis of symmetry and geometric similarity, these 256 cases can ultimately be distilled into 15 fundamental topological structures[8], as illustrated in Figure 1. The system traverses the entire dataset, connecting the triangular facets generated by each voxel to ultimately complete the three-dimensional reconstruction. By iterating through each voxel within the three-dimensional scalar field, the topological structure of the isosurface within the voxel is determined based on the relative positional relationship between the voxel vertices and the preset isosurface threshold.

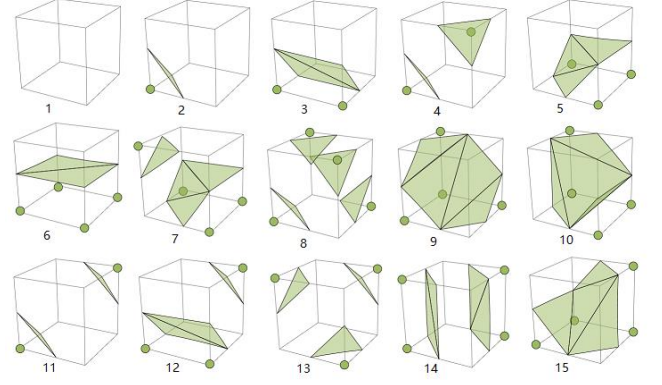


Fig. 1. The 15 topological structures of the traditional MC algorithm

The traditional MC algorithm generates isosurfaces by traversing all voxels. However, due to the vast amount of voxel data and the presence of a large number of empty voxels that do not contain isosurfaces, significant time is wasted during the computation process. Moreover, the MC algorithm processes each voxel independently during computation, without fully considering the positional relationships between adjacent voxels. This oversight can lead to the emergence of holes between neighboring voxels, thereby compromising the integrity and accuracy of the reconstructed model. As illustrated in Figure 2, the yellow rectangular areas represent the missing hole. It is known that the external contours of any object are continuous surfaces, and this holds true for 3D reconstruction as well. Therefore, the isosurfaces generated by adjacent voxels are not independent entities but rather interconnected parts of a whole.

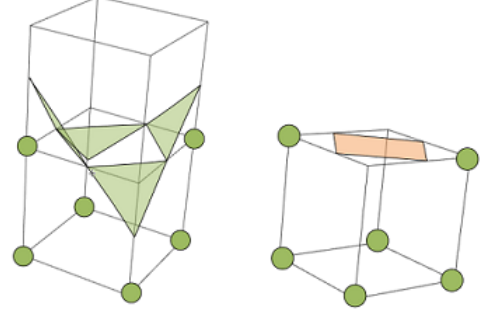


Fig. 2. Example of a hole generated due to ambiguity

To address the aforementioned issues, this paper proposes an improved MC algorithm. This algorithm leverages the existing isosurfaces to form shared edges in adjacent voxels and, based on the theory of surface connectivity, generates new isosurfaces in neighboring voxels[9]. This method not only prevents the occurrence of holes but also enhances the integrity and accuracy of the reconstructed model. As shown in Figure 3, two voxels share a common face ABCD. If the four vertices are not simultaneously greater than or less than the given value, then the adjacent voxel sharing an edge with the seed triangle also contains an isosurface. The line segment EF is the intersection line of the seed triangle on the common face. Since this face also belongs to the voxel on the right, the isosurface within the right voxel must intersect this face at the same intersection line. This proves that the voxel on the right also contains an isosurface.

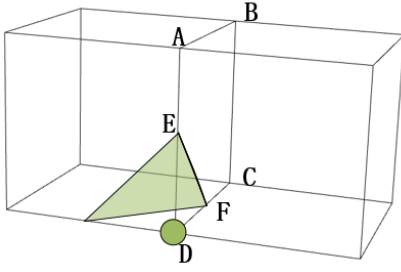


Fig. 3. Example of isosurface intersection line in adjacent voxels

Each voxel has six faces: front, back, top, bottom, left, and right. By determining the state (above or below the threshold) of the four vertices on each face, one can ascertain whether there is an isosurface intersection line. There are four scenarios for the vertex states: one vertex differs from the others in one scenario, two vertices differ from the others in two scenarios, and in the remaining cases, all vertices share the same state, as depicted in Figure 4. If any of the three states 1, 2, or 3 are present, then the voxel with the common face intersection line of the seed triangle contains an isosurface.

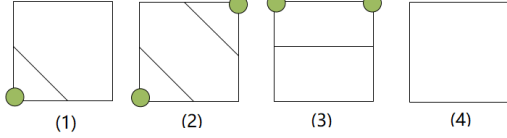


Fig. 4. Intersection line conditions of each face with the seed triangle

The algorithm proposed in this paper employs a connectivity queue and a hash table to assist in representing the entire reconstruction process. The connectivity queue stores information about the seed triangles and the intersection lines of adjacent voxels, while the hash table contains the state and coordinates of each voxel vertex as well as related information about the seed triangles. The implementation steps of the algorithm are as follows:

- 1) For papers with more than six authors: Add author names horizontally, moving to a third row if needed for more than 8 authors.
- 2) Retrieve the seed triangle and intersection line information from the connectivity queue, interpolate the voxel edges according to the topological structure, and connect the intersection points to form new triangular isosurfaces.
- 3) Place the information of the triangles formed in step 2 and the intersection lines of adjacent voxels into the queue, while also adding the seed triangle and intersection line information along with the corresponding voxel vertex states and coordinate information into the hash table.
- 4) Repeat steps 2 and 3 until the queue is empty, at which point the algorithm terminates.

In the selection of seed triangles, our criterion is to choose triangles that are structurally simple and occupy a large proportion as seed triangles. This selection not only simplifies the subsequent processing but also enhances the overall efficiency and stability of the algorithm. In Figure 4, we select the topological structure 1 as the seed triangle, and then, based on the theory of isosurface connectivity, we connect the isosurfaces of adjacent voxels [10]. Upon completion of the algorithm process, the voxel information is retrieved from the hash table to calculate the isosurface points and normal vectors.

The traditional formulas for calculating isosurface points and normal vectors are as follows:

$$p = p_1 + \frac{(isoValue - v_1)(p_2 - p_1)}{v_2 - v_1} \quad (1)$$

$$n = n_1 + \frac{(isoValue - v_1)(n_2 - n_1)}{v_2 - v_1} \quad (2)$$

Here, p represents the coordinates of the isosurface point; p_1 and p_2 are the coordinates of any two endpoints; $isoValue$ is the threshold; v_1 and v_2 denote the weight values of the two endpoints; n_1 and n_2 represent the normal vectors of the selected two endpoints. The traditional linear interpolation method can result in the generation of many very small triangles when the isosurface closely passes through the nodes. To circumvent the creation of such minuscule triangles, this paper introduces the midpoint selection method for interpolation reconstruction. Instead of calculating the vertices of the triangular isosurfaces through linear interpolation as in the conventional MC algorithm, the vertices are determined by using the midpoint of the edge on which the vertex lies. By utilizing the midpoint to replace the vertices of the triangular isosurfaces, the interpolation calculation step can be substituted, thereby reducing computation time. Assuming p_1 and p_2 are two vertices on one side of a voxel and p represents the midpoint, the midpoint selection method yields:

$$p = (p_1 + p_2) / 2 \quad (3)$$

If n_1 and n_2 are the normal vectors of p_1 and p_2 respectively, then the normal vector n of p is given by:

$$n = (n_1 + n_2) / 2 \quad (4)$$

The advantages of midpoint selection are:

- 1) The error precision range is within 0.5 voxel edge length, resulting in minimal visual differences compared to the reconstruction from the traditional MC algorithm.
- 2) If the coordinate values are first scaled up to eliminate decimals, integer operations can be performed, avoiding the need for floating-point calculations.
- 3) It can reduce the computational load, shorten the computation time, and simultaneously prevent the formation of minuscule triangles, decrease the number of closely connected nodes within adjacent voxels, and make the local surface more even, as illustrated in Figure 5.

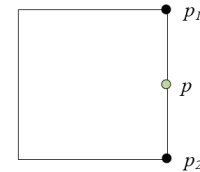


Fig. 5. Schematic diagram of the midpoint selection method

III. CONSTRUCTION OF THREE-DIMENSIONAL SCALAR FIELD DATA

The data utilized in this study originates from DICOM files of a patient from a Beijing hospital, comprising 98 CT images and one RTSTRUCT file. These CT images capture the upper body region in a head-first supine (HFS) position, with each image boasting a resolution of 512×512 pixels and a pixel spacing of 5mm, ensuring high-precision spatial localization. Each pixel is represented by 16 bits, providing a high grayscale resolution that allows for the capture of subtle tissue variations.

The grayscale values typically range from -1024 to +3071, corresponding to different tissue densities. For instance, the grayscale value for air is close to -1000, water is approximately 0, and bone can exceed +1000. The RTSTRUCT file contains manually or automatically delineated contour information for the tumor target area and other critical structures, used for precisely defining the target regions in the treatment plan.

When employing the MC algorithm for three-dimensional data reconstruction, the initial step is to perform scalar quantization on the data. Specifically, this paper begins by reading in the CT image sequence and creating a scalar field data structure of the same size as the CT images (with dimensions rows $\times$ cols $\times$ slices) [11]. Subsequently, based on the image slice thickness, pixel spacing, and the coordinate position of the upper-left corner of the images, the physical coordinates corresponding to each voxel in the scalar field are calculated. The three-dimensional scalar field model is illustrated in Figure 6.

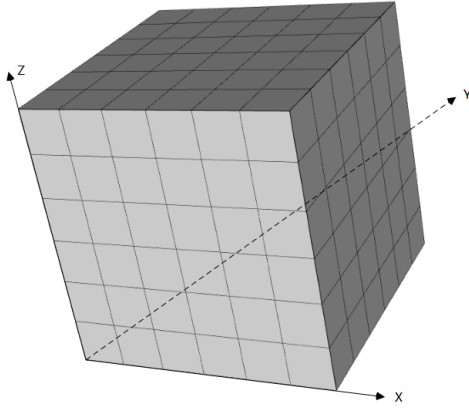


Fig. 6. Three-dimensional scalar field model

Then, the positions of the contour points are read from the DICOM-RTSTRUCT file and mapped into the aforementioned three-dimensional scalar data field. The values inside the contours are assigned as 1, while others are set to 0 [12]. Initially, the physical coordinates of the contour points are converted into the indices (i, j, k) of each voxel in the image, as shown in Figure 7. The image coordinate system describes how to acquire images of the relevant anatomical structures. Taking CT (Computed Tomography) as an example, the scanner creates a regular rectangular grid of points and cells, which are arranged starting from the upper-left corner. The origin of the coordinate system is located at the upper-left corner, with the X-axis extending to the right, the Y-axis downward, and the Z-axis along the patient's body. The center of each voxel serves as a point, arranged at regular intervals to form a three-dimensional array of points, with adjacent eight points constituting a cubic cell. This regular structure provides a foundation for subsequent image processing and three-dimensional reconstruction. Here, i, j, k correspond to the X-axis, Y-axis, and Z-axis, respectively. The origin represents the first pixel in the anatomical coordinate system, and the image spacing indicates the distance between voxels on each axis. Using the origin and spacing, the index positions corresponding to the contour points can be calculated. Then, based on the calculated index values, the corresponding voxels are located and assigned a value of 1. Finally, all points within the entire contour are filled and assigned a value of 1, thereby completing the marking of the region of interest.

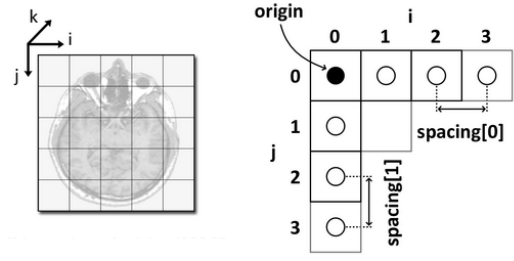


Fig. 7. Image coordinate system

IV. VISUALIZATION SYSTEM AND EXPERIMENTAL RESULTS

The experimental environment of this paper is shown in Table I. The experiment utilized the improved MC algorithm proposed in this paper, combined with the Visualization Toolkit (VTK) [13] for data visualization and analysis, achieving three-dimensional visualization processing of the target area data.

TABLE I. EXPERIMENTAL ENVIRONMENT

Hardware & Software	Version Configuration
Operating system	Windows 10
CPU	Intel(R) Core(TM) i5-4590 CPU
GPU	NVIDIA GeForce RTX 2080 Ti
RAM	32 GB DDR4
Storage	1 TB NVMe SSD
VTK	9.0.1

VTK, as a globally leading underlying visualization toolkit, integrates a wealth of visualization algorithms and advanced modeling techniques. Its functionalities span across multiple domains including two-dimensional plotting, three-dimensional computer graphics, image processing, volume rendering, and complex modeling. The toolkit significantly enhances system operational efficiency and scalability through optimized thread management and distributed memory parallel processing technologies [14]. To comprehensively evaluate the algorithm's performance, we selected three representative delineation data from the RTSTRUCT file, corresponding to the body's external contour, the left lung, and the esophagus. The delineation data for these three parts exhibit a distribution characteristic from more to fewer contour points, which can adequately reflect the processing effects of contours with different complexities. The specific data volume of contour points for each part and their original delineation schematic diagrams are shown in Table II.

TABLE II. THREE GROUPS OF DIFFERENT NUMBERS OF CONTOUR POINT DATA SOURCE INFORMATION

Organ	Schematic diagram	Points
External		33993
Left Lung		5335
Esophagus		2153

This paper focuses on creating corresponding scalar field data for three parts: the body's external contour, the left lung, and the esophagus. Initially, both the traditional MC algorithm and the improved MC algorithm were employed to perform surface three-dimensional reconstruction to compare the performance differences between the two algorithms. The triangular isosurface patches generated during the reconstruction process serve as the foundational data source and are imported into the vtkPolyData data structure. During the visualization processing stage, vtkActor and vtkMapper objects are created and bound, with the output data from vtkPolyData being passed to vtkMapper. Ultimately, by adding vtkActor to vtkFullScreenRenderWindow for rendering, the three-dimensional reconstruction and visual display of the target area are achieved [15]. The comparison of reconstruction effects is shown in Figure 10.

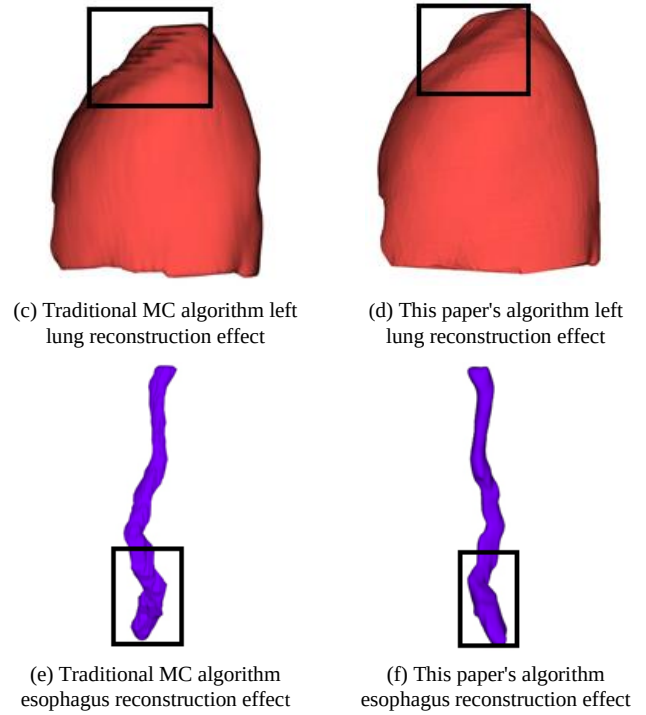
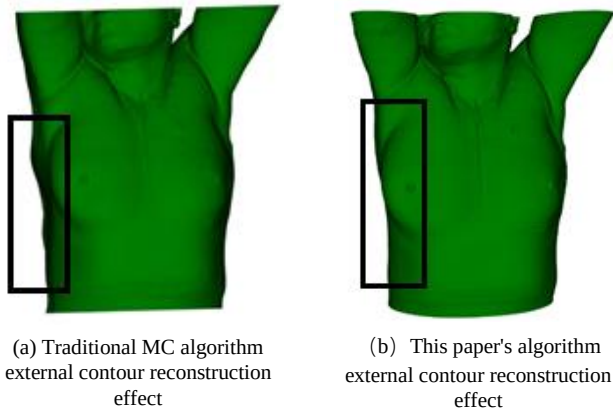


Fig. 8. Comparison of reconstruction effects for three sets of data

From the comparative results in Figure 10, it is evident that the improved MC algorithm produces a smoother reconstructed surface compared to the traditional algorithm when dealing with inflection points or areas with significant data morphology turns, better maintaining the continuity of the contours. To further quantify the algorithm's performance, this study conducted a statistical analysis of the computation time and the number of triangular isosurfaces generated by the two reconstruction methods, with specific comparative results presented in Table III and Figure 11.

TABLE III. COMPARISON OF THE 3D RECONSTRUCTION RESULTS OF THE THREE SETS OF DATA IN THE EXPERIMENT

Organ	Algorithm	Time (ms)	Triangle faces
External	MC	1948.2	112594
	Ours	1676.5	104478
Left Lung	MC	342.8	16984
	Ours	292.8	16124
Esophagus	MC	220.5	6972
	Ours	185.1	6651

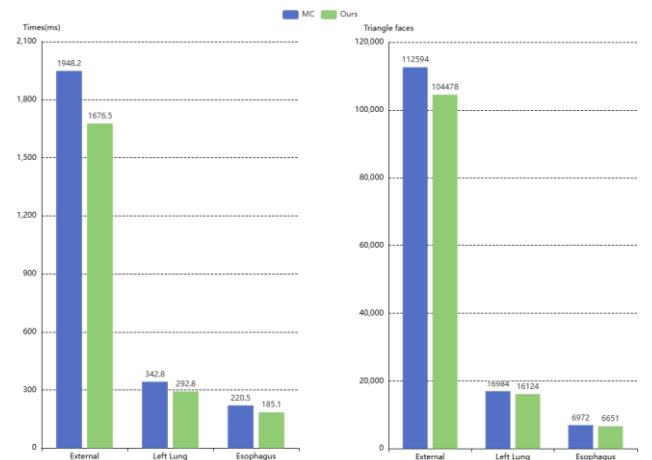


Fig. 9. Comparative Diagram of Reconstruction Results for Three Sets of Data

Based on the comparative analysis of experimental data from Table 3 and Figure 11, it is evident that the improved MC algorithm has achieved significant performance enhancements: the reconstruction time is reduced by more than 16% compared to the traditional MC algorithm, and the number of triangular isosurface patches generated is decreased by over 4%. These results indicate that the improved algorithm not only enhances computational efficiency but also optimizes the quality of isosurfaces, reducing redundant data while ensuring reconstruction accuracy. Considering multiple indicators such as reconstruction quality, computational efficiency, and isosurface optimization, the improved algorithm proposed in this paper demonstrates superior overall performance to the traditional MC algorithm.

V. CONCLUSION

This paper has realized the three-dimensional reconstruction technology of tumor target areas by applying an improved MC algorithm. By introducing the triangle surface growth connectivity theory and combining it with the innovative design of the midpoint selection method to replace the traditional linear interpolation method, the three-dimensional reconstruction task has been efficiently completed, achieving remarkable results in experiments. The experiments demonstrate that the utilization of this paper's algorithm can not only reduce the reconstruction time by more than 16% but also decrease the extraction of triangular isosurface patches by over 4%, discarding those triangular patches that do not substantially enhance the model's accuracy. Under the premise of ensuring model integrity, the number of triangular patches has been streamlined. This optimization not only further reduces the computational load but also results in a smaller amount of data for the reconstructed models, making them easier for subsequent processing and application. Moreover, the reconstruction effect is smoother, with more delicate details, effectively avoiding the jagged or uneven phenomena that may occur with traditional algorithms. Adopting the algorithm proposed in this paper, the three-dimensional reconstruction speed of the target area is significantly improved, and the generated models have clear boundaries and high fidelity in texture details. This enables doctors to formulate radiotherapy plans more efficiently and accurately, thereby enhancing treatment efficacy and improving patients' quality of life. However, the current algorithm still has the issue of the external contour not being smooth enough, where the composition of triangular patches can be clearly observed. This identifies the key direction for future research, which is to conduct specialized smoothing processing on the external contours after three-dimensional reconstruction to further enhance model precision, meeting the growing demand for high-precision three-dimensional reconstruction models in fields such as medical imaging, computer graphics, and virtual reality.

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